SCHOLAR-PATH

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report **"SCHOLAR-PATH"** is the bonafide work of **"IKSHITA (23BCS13864), DIVYA (23BCS13027), PARUL NANDA (23BCS13044), ANU (23BCS14326)"** who carried out the project work under my supervision.

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ABSTRACT

The ScholarPath is a web-based application developed using Java and Spring Boot to automate and streamline the complete process of scholarship distribution in educational institutions. The system replaces traditional manual processes with a secure, efficient, and centralized digital platform.

The application supports three main user roles: Student, Verifier, and Administrator. Students can register, log in, apply for scholarships, upload required documents, and track application status. Verifiers review applications and validate documents, while Administrators make final decisions regarding approval or rejection.

The backend is developed using Spring Boot with a layered architecture (Controller, Service, Repository), while the database is managed using MySQL with Spring Data JPA (Hibernate). The system implements JWT-based authentication and role-based access control using Spring Security to ensure secure communication and restricted access.

Additional features include file upload handling, help request escalation, and document retrieval APIs. The system improves transparency, reduces processing time, and enhances data security. This report presents the complete development lifecycle including design, implementation, testing, and future scope.

GRAPHICAL ABSTRACT

ScholarPath – Workflow Overview

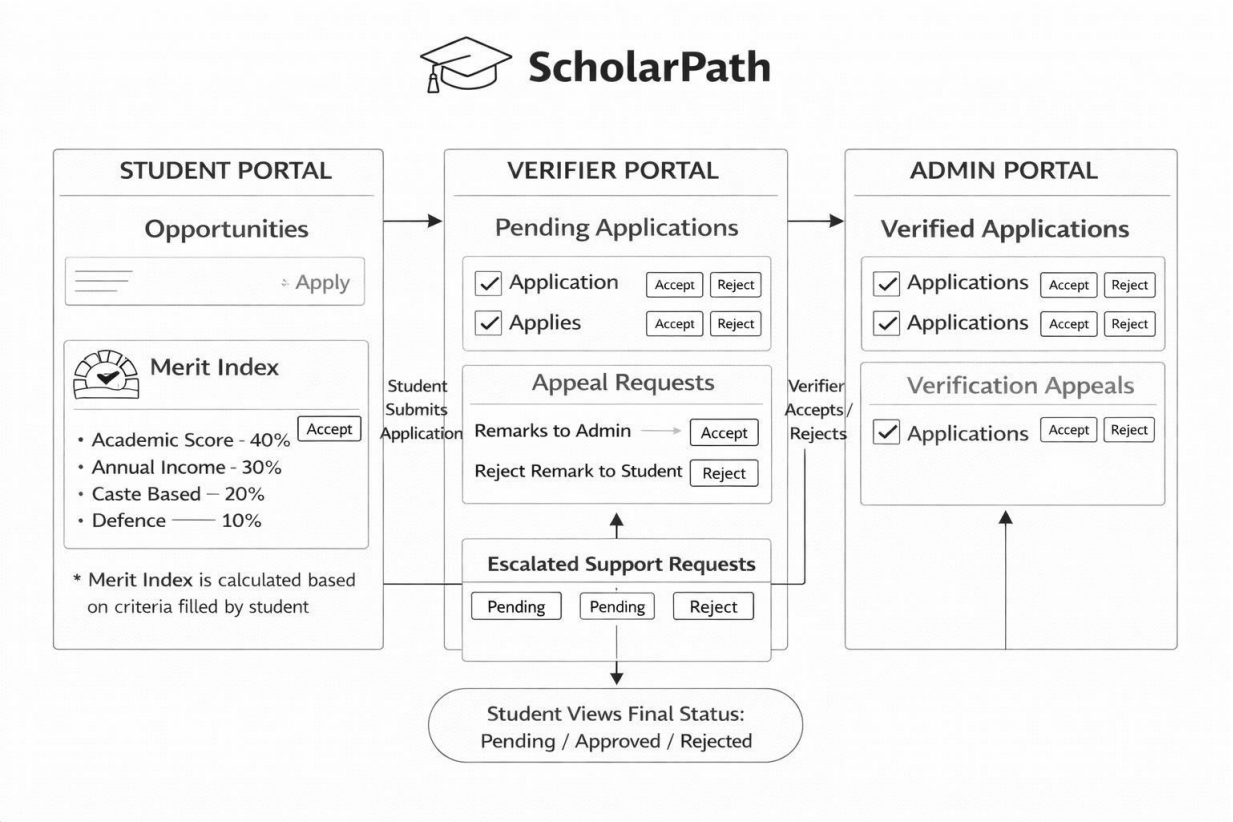


Figure G.1: Graphical Abstract of the Scholarship Management System Workflow

ABBREVIATIONS AND SYMBOLS

Abbreviation	Full Form
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
SQL	Structured Query Language
MySQL	MySQL Server
UI	User Interface
UX	User Experience
MVC	Model-View-Controller
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity-Relationship
DB	Database
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
OTP	One-Time Password
PDF	Portable Document Format

CHAPTER 1

INTRODUCTION

1.1 Identification of Client and Need

In the contemporary educational landscape, scholarship programs play a vital role in providing financial assistance to deserving students. Educational institutions, government agencies, and private organizations regularly run scholarship programs that cater to students based on merit, income level, category, and other criteria. The primary clients for the Scholarship Management System include universities, colleges, state government education departments, and NGOs that administer scholarship programs.

The need for an automated scholarship management system arises from the increasing volume of applications received every academic year. Managing thousands of applications manually is highly error-prone and time-consuming. Students often have difficulty navigating complex paper-based processes, resulting in missed opportunities due to incomplete submissions or missed deadlines. Institutions, on the other hand, struggle to maintain organized records and ensure transparent decision-making.

The ScholarPath addresses this need by providing a centralized, secure, and user-friendly web-based platform where students can apply online, administrators can manage and process applications efficiently, and the entire scholarship lifecycle can be tracked in real time. The system eliminates geographical barriers, reduces paperwork, and makes the scholarship process fair, transparent, and accessible to all eligible students.

1.2 Identification of Problem

The traditional approach to scholarship management is heavily dependent on manual procedures. Students are required to collect application forms physically, fill them by hand, attach physical copies of documents, and submit them in person. This approach leads to numerous issues that negatively affect both students and administrators:

- Time-Consuming Process: Manual review and verification of hundreds or thousands of scholarship applications demands significant human effort and time, often resulting in delays in announcing results.

- Data Inaccuracy: Handwritten forms are prone to illegible entries, errors in data recording, and loss of information due to physical damage or misplacement of documents.
- Lack of Transparency: In a manual system, students have no way to track the progress of their application. There is often no clear communication regarding approval or rejection, leading to confusion and dissatisfaction.
- Duplication of Effort: Without a centralized system, the same student data is often entered multiple times into different registers or spreadsheets, wasting time and increasing the risk of inconsistencies.
- Security Concerns: Physical documents are susceptible to theft, damage, and unauthorized access. Sensitive student data such as income certificates and caste certificates may be mishandled.
- Administrative Burden: Staff members spend excessive time in sorting, categorizing, and reviewing applications instead of focusing on value-added tasks such as policymaking and student welfare.

These problems collectively demand a robust digital solution that automates the scholarship process, ensures data security, and improves the overall experience for both students and administrators.

1.3 Identification of Tasks

To develop a fully functional Scholarship Management System, the following tasks were systematically identified and executed:

- **Requirements Analysis:** Identifying functional and non-functional requirements, understanding user roles (Student, Verifier, Administrator), and documenting system use cases.
- **System Architecture Design:** Designing the overall structure of the application using a layered architecture (Controller–Service–Repository) in Spring Boot to ensure separation of concerns, scalability, and maintainability.
- **Database Design:** Creating an Entity-Relationship (ER) diagram and designing normalized database tables in MySQL to efficiently store user profiles, application data, documents, and decision records.

- **Frontend Development:** Developing responsive and user-friendly web pages using HTML5, CSS3, and JavaScript (React) to ensure seamless interaction across different devices and browsers.
- **Backend Development:** Building server-side logic using Spring Boot (Java) to handle business rules, authentication, JWT-based security, and application processing.
- **Role-Based Access Control:** Implementing role-based access using Spring Security with roles such as Student, Verifier, and Administrator to ensure secure and controlled access to system functionalities.
- **Document Upload Module:** Enabling students to securely upload supporting documents such as income certificates, mark sheets, and ID proofs using multipart file handling.
- **Eligibility Verification Module:** Implementing business logic in the service layer to evaluate applications based on criteria such as academic performance, income, caste, and other factors.
- **Testing and Validation:** Conducting unit testing, integration testing, and user acceptance testing to ensure the system performs reliably under various conditions.
- **Documentation:** Preparing a detailed project report and user manual explaining system features, workflow, and usage instructions.

1.4 Timeline

The project was planned and executed over a period of eight weeks. A structured timeline was followed to ensure timely completion of all tasks. The development phases and their corresponding timelines are described below:

Week	Phase	Activities
Week 1	Requirements Gathering	Problem definition, client need analysis, and documentation of system requirements
Week 2	System and DB Design	ER diagram, database schema creation, MVC architecture planning

Week 3-4	Backend Development	Spring Boot java coding for authentication, application processing, and admin logic
Week 5	Frontend Development	HTML/CSS/JS (react)pages for student, verifier, and admin interfaces
Week 6	Integration	Connecting frontend with backend APIs and database
Week 7	Testing & Validation	Unit tests, integration tests, bug fixes, and user acceptance testing
Week 8	Documentation	Final report writing, user manual preparation, and project submission

Table 1.1: Project Timeline

Figure 1.1 illustrates the Gantt chart representation of the project timeline showing the week-by-week breakdown of all development phases.

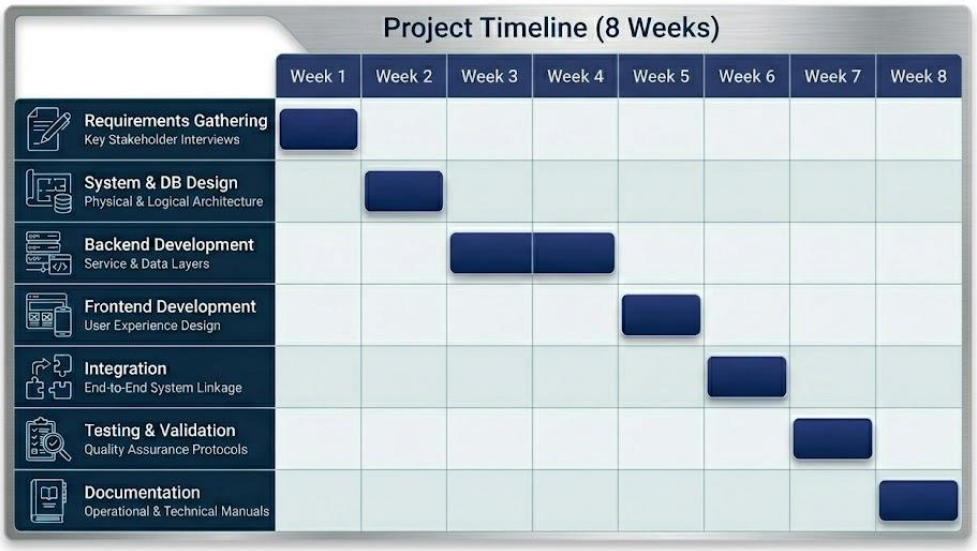


Figure 1.1: Gantt Chart – Project Timeline

1.5 Organization of the Report

This project report is organized into five chapters, each addressing a specific aspect of the development process:

- Chapter 1 – Introduction: Presents the project background, identification of client need, problem statement, tasks involved, project timeline, and report structure.
- Chapter 2 – Literature Survey: Reviews existing scholarship and application management systems, presents bibliometric analysis, summarizes related work, and defines the problem and objectives clearly.
- Chapter 3 – Design Flow / Process: Describes concept generation, feature evaluation, design constraints, alternative design approaches, final design selection, and the complete system flowchart.
- Chapter 4 – Results Analysis and Validation: Covers the implementation using modern tools, showcases system screenshots, presents test cases, and discusses performance metrics and validation results.
- Chapter 5 – Conclusion and Future Work: Summarizes the project achievements, discusses advantages and limitations, outlines real-world applications, presents the conclusion, and proposes future improvements.

CHAPTER 2

LITERATURE SURVEY

2.1 Timeline of the Reported Problem

The challenge of managing scholarship applications efficiently has evolved significantly over several decades. In the early era of educational administration, all scholarship processes were handled through paper-based systems. Students were required to physically collect forms, fill them manually, attach photocopies of documents, and submit them to administrative offices. This process was not only time-consuming but also extremely prone to human error and document misplacement.

During the 1990s, as personal computers began to proliferate in educational institutions, some organizations started maintaining digital records using spreadsheet software such as Microsoft Excel. While this represented a step forward in data management, it still required manual data entry and did not provide any means for students to interact with the system digitally.

The early 2000s saw the emergence of basic web-based systems that allowed students to submit information online through static HTML forms. However, these early systems lacked dynamic functionality, real-time validation, and secure backend processing. Data was often stored in flat files rather than structured databases, making retrieval and analysis cumbersome.

From 2005 onwards, with the rapid growth of web technologies, database management systems, and internet accessibility, more sophisticated online application systems began to appear. Government bodies in countries like India, the United States, and European nations started developing dedicated scholarship portals. In India, the National Scholarship Portal (NSP) was launched by the Ministry of Electronics and Information Technology (MeitY) to consolidate various scholarship schemes onto a single platform.

Despite these advancements, many institutional-level scholarship systems remain fragmented, manually operated, or poorly integrated. The need for a comprehensive, role-based, institution-specific scholarship management system remains a relevant and important engineering challenge as of 2026.

2.2 Bibliometric Analysis

A review of published research papers, conference proceedings, and technical reports over the past decade reveals a growing interest in the development of digital application and verification systems for academic institutions. The key focus areas identified from the literature are as follows:

- Web-Based Application Systems: A majority of research papers focus on developing browser-based platforms for managing institutional processes, emphasizing the importance of usability, responsiveness, and cross-platform compatibility.
- Database Management and Security: Studies consistently highlight the importance of using relational database management systems (RDBMS) for structured data storage and emphasize the need for encryption and access control to protect sensitive student data.
- Role-Based Access Control (RBAC): Multiple researchers have demonstrated that implementing RBAC significantly enhances system security and workflow management in multi-user environments.
- Automated Eligibility Verification: Recent works propose automation of eligibility checks using predefined rule engines, reducing the need for manual evaluation and minimizing subjective bias.
- Mobile-Responsive Design: Emerging research stresses the importance of mobile-first design in student-facing systems, given the widespread use of smartphones among college students.

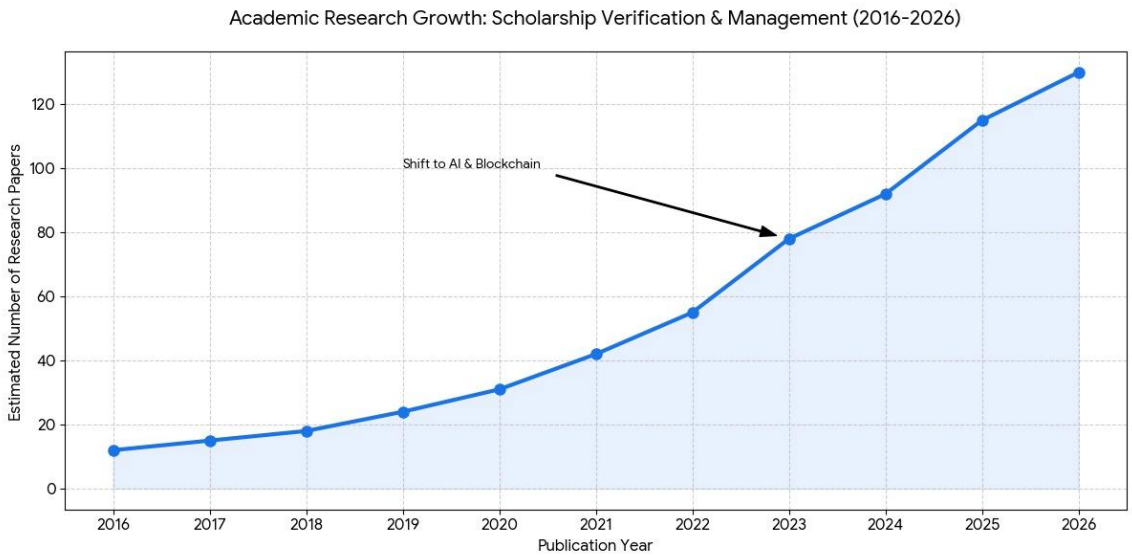


Figure 2.1: Bibliometric Analysis Graph – Research Focus Areas

2.3 Proposed Solutions by Different Researchers

Various researchers and developers have proposed different solutions to address the challenges in application and scholarship management. A review of their contributions is presented below:

2.3.1 Online Form-Based Systems

Kumar and Sharma (2018) proposed a web-based student application system for a state university that allowed students to fill and submit forms online. The system used PHP for backend processing and MySQL for database management. While the system successfully reduced paper usage, it lacked document upload capability and real-time status tracking, limiting its practical usefulness.

2.3.2 Document Verification Systems

Patel et al. (2019) developed a digital document verification module integrated with a university administration system. Their system used file upload functionality and admin review panels to verify submitted documents. However, the system did not support role-based access, meaning all administrators had access to all types of applications regardless of their responsibility area.

2.3.3 Automated Eligibility Checking

Mehta and Joshi (2020) introduced an automated scholarship eligibility checker that used predefined criteria stored in a configuration file. The system automatically filtered applications based on academic and financial parameters. This approach reduced manual

review time significantly but required periodic updates to the criteria file, which was a manual and error-prone process.

2.3.4 AI-Based Fraud Detection

Some advanced research works, such as those by Chen et al. (2021), have explored the use of machine learning algorithms to detect fraudulent documents submitted during scholarship applications. While promising, these approaches require large training datasets and significant computational resources, making them impractical for smaller institutions with limited IT infrastructure.

2.3.5 National-Level Scholarship Portals

Government-developed portals such as India's National Scholarship Portal (NSP) and the United States' Federal Student Aid (FAFSA) system serve millions of students. These systems are comprehensive but are designed for national-scale operations. They often lack the customization needed for institution-specific requirements, local eligibility criteria, and departmental-level management.

Author / System	Technology	Key Features	Limitation
Kumar & Sharma (2018)	PHP, MySQL	Online forms, reduced paperwork	No document upload
Patel et al. (2019)	ASP.NET, SQL Server	Document upload, admin review	No RBAC
Mehta & Joshi (2020)	Java, MySQL	Automated eligibility check	Manual criteria updates
Chen et al. (2021)	Python, ML	AI fraud detection	High resource needs
NSP, FAFSA (Govt.)	Various	Large-scale, comprehensive	No customization

Table 2.1: Summary of Literature Review

2.4 Summary and Problem Definition

From the extensive literature survey conducted, it is evident that while numerous solutions exist for online application and scholarship management, significant gaps remain in terms of complete automation, institution-level customization, secure role-based access, and integrated verification workflows. Most existing systems address only one or two aspects of the scholarship lifecycle, leaving the rest to manual processing.

The problem can be formally defined as follows: Educational institutions lack a fully integrated, secure, and automated scholarship management system that supports the complete application lifecycle from student registration and form submission to document verification, eligibility checking, and final decision communication — all within a single, role-based, web-accessible platform.

The proposed Scholarship Management System aims to bridge this gap by delivering a complete end-to-end solution tailored for use by individual educational institutions.

2.5 Goals and Objectives

Based on the problem definition and the gaps identified in the literature, the following goals and objectives have been established for this project:

- To develop a fully functional, web-based Scholarship Management System accessible through standard browsers without the need for any additional software installation.
- To implement a secure user registration and authentication system with role-based access control for Students, Verifiers, and Administrators.
- To create an intuitive and responsive student interface for completing scholarship application forms and uploading required documents.
- To build a verifier module that enables efficient review and validation of submitted documents, with the ability to forward or reject applications with remarks.
- To develop an administrator dashboard for final evaluation of applications, management of scholarship records, and generation of reports.
- To automate the eligibility check process based on predefined criteria such as minimum academic percentage and maximum family income threshold.
- To provide real-time status tracking for students, allowing them to monitor their application progress at any point.
- To ensure data security, integrity, and privacy through secure connections, encrypted passwords, and controlled data access.
- To reduce the overall processing time and administrative workload associated with scholarship management.

CHAPTER 3

DESIGN FLOW / PROCESS

3.1 Concept Generation

The concept for the Scholarship Management System was generated through a systematic analysis of the existing scholarship workflow at educational institutions. The development team brainstormed various approaches to digitize and automate the process.
The initial brainstorming sessions identified the following key concepts:

- Centralized Portal: A single web-based platform where all scholarship-related activities — registration, application, verification, and approval — take place within a unified interface.
- Role-Differentiated Access: A system that presents different interfaces and functionalities based on the type of user (Student, Verifier, Administrator) logged into the system.
- Document-Centric Verification: A workflow where document upload and verification form the core of the scholarship evaluation process, ensuring that all claims made by applicants are supported by authentic documents.
- Automated Eligibility Filtering: A rule-based engine that automatically evaluates applications against predefined criteria before passing them on to human reviewers, reducing unnecessary manual workload.
- Real-Time Status Communication: A mechanism by which students can check the current status of their application (Pending, Under Review, Verified, Approved, or Rejected) at any time through their personal dashboard.

After evaluating these concepts, the team decided to combine all elements into a single cohesive system architecture that supports the complete scholarship management lifecycle within one integrated platform.

3.2 Evaluation and Selection of Specifications / Features

Each feature was evaluated on the basis of importance, feasibility, and alignment with user needs. The following feature evaluation matrix was used to finalize the system specifications:

Feature	Importance (1-5)	Feasibility (1-5)	User Impact (1-5)	Selected?
User Registration & Login	5	5	5	Yes
Role-Based Access Control	5	5	5	Yes
Online Application Form	5	5	5	Yes
Document Upload	5	4	5	Yes
Eligibility Check	4	4	4	Yes
Verifier Review Panel	5	5	4	Yes
Admin Dashboard	5	5	5	Yes
Real-Time Status Tracking	4	4	5	Yes

Table 3.1: Feature Evaluation and Selection Matrix

3.3 Design Constraints

The design of the Scholarship Management System was subject to several constraints across different dimensions. These constraints were identified early in the design phase and appropriate mitigation strategies were planned:

Constraint Type	Constraint	Mitigation Strategy
Technical	Backend framework limitation (Spring Boot architecture)	Follow layered architecture (Controller–Service–Repository) for maintainability
Security	Data privacy of students	Implement HTTPS, hashed passwords, session management
Organizational	Defined workflow hierarchy (Student → Verifier → Admin) must be maintained	Enforce role-based access control and structured approval flow
Operational	System usability for different types of users (Students, Verifiers, Admins	Design a simple and intuitive user interface with clear workflows
Safety	Data loss prevention	Implement regular DB backups and error handling

Table 3.2: Design Constraints and Mitigation Strategies

3.4 Alternative Designs

During the design phase of the ScholarPath system, different architectural approaches were considered to ensure scalability, maintainability, and efficient workflow management.

Alternative Design 1 – Monolithic Architecture

In this approach, the entire application, including frontend, backend logic, and database interaction, is developed and deployed as a single unified system. All functionalities are tightly integrated, and there is no clear separation between different layers of the application.

Advantages:

1. Simple to develop and deploy
2. Minimal configuration required
3. Suitable for small-scale applications.

Disadvantages:

1. Poor separation of concerns
2. Difficult to maintain and scale
3. Tightly coupled components make testing and updates complex.

Alternative Design 2 – Layered Architecture using Spring Boot (Selected)

In this approach, the application is structured using a layered architecture consisting of Controller, Service, and Repository layers. The Controller layer handles REST API requests, the Service layer manages business logic such as application processing and eligibility evaluation, and the Repository layer interacts with the database using Spring Data JPA.

This design also supports integration with a separate frontend (React), enabling a clear separation between client-side and server-side components.

Advantages:

1. Better separation of concerns; improved maintainability
2. Easier testing and scalability
3. Supports RESTful APIs and modern web development practices.

Disadvantages:

1. Slightly higher initial complexity
2. Requires proper design and configuration.

3.5 Best Design Selection

After a thorough evaluation of both alternative designs, the Modular MVC Architecture (Alternative Design 2) was selected as the best design for the Scholarship Management System. The following comparison table presents the rationale for this selection:

Criterion	Design 1 (Monolithic)	Design 2 (MVC) – Selected
Code Maintainability	Low – tightly coupled code	High – clean separation of layers
Scalability	Poor – difficult to scale modules independently	Good – modules can scale independently
Testability	Difficult – UI and logic mixed	Easy – each layer tested independently
Security	Average	Strong – centralized controller with middleware
Industry Standard	Less common in modern apps	Widely used and accepted standard
Development Time	Faster initially	Slightly longer but more structured
Long-term Suitability	Low	High

Table 3.3: Comparison of Alternative Designs

The MVC architecture was unanimously selected because it provides a well-organized, maintainable, and scalable foundation for the application. The separation of concerns ensures that the UI do not affect the business logic

3.6 Implementation Plan – Flowchart and Algorithm

The following flowchart represents the complete workflow of the ScholarPath system, starting from user authentication to final decision-making.

The workflow consists of the following steps:

- The student accesses the system and registers or logs in using valid credentials.
- After successful authentication, the student navigates to the dashboard and selects available scholarship opportunities.
- The student fills out the application form with required details such as academic performance, family income, and other relevant information.

• The student uploads supporting documents (e.g., income certificate, mark sheets, ID proof) using the file upload module.

The system processes the application and evaluates it based on predefined criteria. The application is then stored in the database with a status of "Pending".

• The application is forwarded to the Verifier portal for review. The Verifier examines the submitted details and documents.

– If valid, the application is marked as "Verified" and forwarded to the Administrator.

– If invalid, the application is marked as "Rejected" with appropriate remarks.

• The Administrator reviews all verified applications. Based on final evaluation:

– The application is "Approved" if it meets the required criteria.

– Otherwise, it is "Rejected" with remarks.

• The student can track the application status (Pending / Verified / Approved / Rejected) through the dashboard.

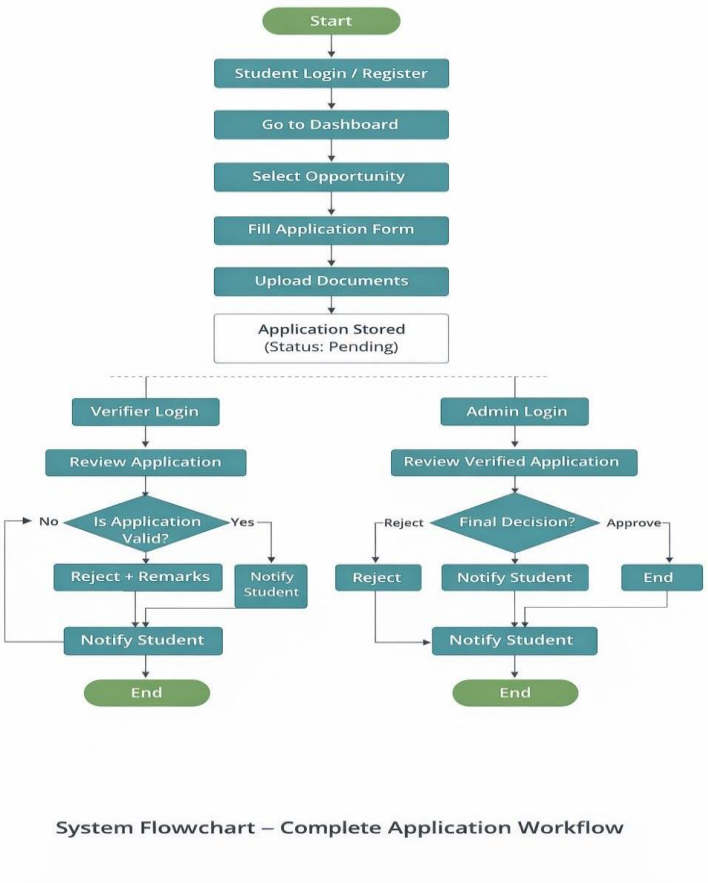


Figure 3.3: System Flowchart – Complete Application Workflow

CHAPTER 4

RESULTS ANALYSIS AND VALIDATION

4.1 Implementation Using Modern Engineering Tools

The ScholarPath was implemented using a well-selected combination of modern web development technologies. Each technology was chosen based on its suitability for the task, community support, availability in the academic environment, and alignment with the MVC design pattern selected in Chapter 3.

Layer	Technology	Purpose
Frontend	HTML5, CSS3	Structure and styling of web pages
Frontend	JavaScript (React)	Client-side validation and dynamic interactions
Backend	Spring Boot (Java)	Server-side logic, routing, session management
Database	MySQL	Structured storage and retrieval of all application data
Code editor	VS Code	Development, debugging, and deployment of the application
Version Control	Git / GitHub	Code versioning and collaboration
Browser Testing	Chrome, Firefox, Edge	Cross-browser compatibility verification
Documentation	MS Word	Report and user manual preparation

Table 4.1: Technology Stack Used in the Project

4.2 Implementation of Design Using Modern Engineering Tools

The ScholarPath system has been developed and implemented using various modern engineering tools and technologies across different phases of the project, including analysis, design, development, testing, and documentation.

1.Analysis and Design:

System design was carried out using flowcharts and architectural diagrams to represent the workflow and system structure. The layered architecture (Controller–Service–Repository) was used to ensure proper separation of concerns and maintainability.

2.Development Tools and Technologies:

The backend was developed using Spring Boot (Java), enabling rapid development of RESTful APIs. Spring Data JPA (Hibernate) was used for database interaction, while MySQL was used for structured data storage. Maven was used for dependency management and project build automation.

3.Security Implementation:

Spring Security along with JWT (JSON Web Token) was used to implement authentication and role-based access control for Students, Verifiers, and Administrators.

4.File Handling and Data Processing:

Multipart file handling was implemented to allow secure uploading and storage of user documents. Business logic, such as application processing and status management was handled in the service layer.

5.Testing and Validation:

API endpoints were tested using Postman to verify request-response cycles. Functional testing was performed to ensure the correct workflow from application submission to final approval. Error handling and validation mechanisms were implemented to maintain data integrity.

6.Documentation and Report Preparation:

Project documentation, including system design, flowcharts, and reports, was prepared using MS Word and diagramming tools.

7.Version Control and Collaboration:

Git and GitHub were used for version control, enabling efficient code management and tracking changes throughout development.

Overall, the use of modern tools and frameworks ensured efficient development, secure implementation, and reliable performance of the system.

4.3 Testing and Validation

A comprehensive testing strategy was implemented to ensure the correctness, reliability, and security of the ScholarPath system. Testing was conducted at multiple levels as described below:

1.Unit Testing

Individual components of the system were tested to verify their correctness. Core functionalities such as password encryption, file upload handling, and database operations were validated using different input conditions, including valid, invalid, and edge cases. Basic validation logic was tested during development.

2. Integration Testing

Integration testing was performed to ensure that different modules of the system work together correctly. The interaction between the application submission module, database storage, and verification workflow was tested. The complete flow from student application submission to verifier review and final admin decision was validated.

3.API Testing

All REST API endpoints were tested using Postman to verify correct request handling, response structure, and status codes. Both successful and failure scenarios were tested to ensure robustness.

4.Functional Testing

The system was tested from the user perspective to verify all functionalities. This included student registration, login, application submission, document upload, verification process, and status tracking.

5.Security Testing

Authentication and authorization mechanisms were tested using JWT tokens. Access control was verified to ensure that only authorized roles (Student, Verifier, Admin) can access specific endpoints.

6.File Upload Testing

The document upload feature was tested with different file types and sizes to ensure proper handling, validation, and storage without errors.

7.Validation and Error Handling

Input validation and exception handling mechanisms were tested to ensure that the system responds appropriately to invalid inputs and unexpected conditions.

Overall, the testing process ensured that the system performs reliably, securely, and as expected under various scenarios.

TC #	Test Case	Expected Result	Actual Result	Status
TC 01	Valid user registration	Account created, redirect to login	Account created successfully	PASS
TC 02	Login with invalid password	Error message displayed	Error displayed correctly	PASS
TC 03	Submit form with empty fields	Validation error shown	Validation triggered correctly	PASS
TC 04	Verifier approves documents	Status changed to Verified	Status updated correctly	PASS
TC 05	Admin approves application	Status set to Approved	Approved and logged correctly	PASS
TC 06	Student checks status	Current status visible	Status displayed correctly	PASS
TC 07	SQL injection in login field	Input sanitized, no breach	Input sanitized correctly	PASS

Table 4.2: Test Cases and Results

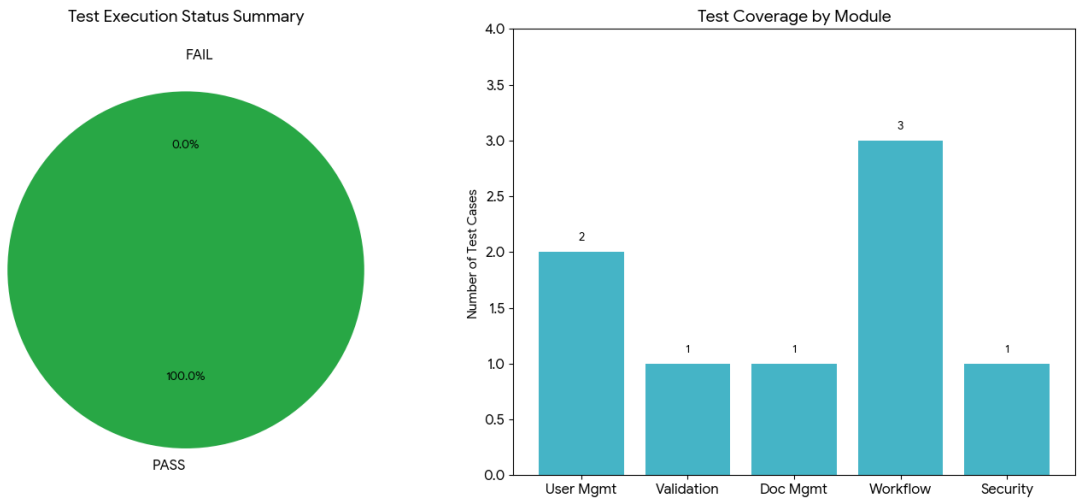


Figure 4.6: Test Case Results Summary

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The Scholar Path system was successfully designed and developed to provide a structured and efficient platform for managing scholarship applications. The system enables students to apply for scholarships, upload required documents, and track application status. It also facilitates a multi-level verification process involving verifiers and administrators, ensuring transparency and accuracy in decision-making.

The use of Spring Boot for backend development, along with Spring Data JPA and MySQL for database management, enabled the creation of a scalable and maintainable system. Security was effectively implemented using JWT-based authentication and role-based access control, ensuring that only authorized users can access specific functionalities.

Overall, the system meets its primary objectives of digitizing the scholarship process, reducing manual effort, and improving workflow efficiency.

5.2 Future Work

The system can be further enhanced with the following improvements:

- Implementation of automated eligibility scoring (Merit Index) for faster decision-making.
- Integration of email/SMS notification services for real-time updates to users.
- Development of a mobile application for improved accessibility.
- Implementation of AI-based document verification and fraud detection.
- Enhanced analytics dashboard for administrators to monitor applications and trends.
- Cloud deployment for better scalability and availability.

Future development will focus on improving automation, enhancing user experience, and scaling the system for real-world deployment. The integration of advanced technologies and cloud infrastructure can transform the system into a fully automated and intelligent scholarship management platform.

REFERENCES

1. Chen, L., Zhang, W. and Liu, R. (2021) 'Machine Learning Approaches for Automated Document Verification in Scholarship Applications', Journal of Educational Technology Systems, Vol. 49, No. 3, pp. 312-329.

2. Government of India, Ministry of Electronics and Information Technology (2016) 'National Scholarship Portal – User Manual', Available at: https://scholarships.gov.in (Accessed: April 2026).

3. Kumar, A. and Sharma, P. (2018) 'Design and Development of an Online Student Application System for State Universities', International Journal of Computer Science and Engineering, Vol. 6, No. 2, pp. 145-152.

4. Mehta, R. and Joshi, S. (2020) 'Automated Eligibility Verification System for Merit-Based Scholarships', Proceedings of the International Conference on Emerging Trends in Engineering, Chandigarh, pp. 87-93.

5. Microsoft Corporation (2023) 'ASP.NET MVC Documentation', Available at: https://docs.microsoft.com/en-us/aspnet/mvc (Accessed: April 2026).

6. Microsoft Corporation (2023) 'SQL Server Documentation', Available at: https://docs.microsoft.com/en-us/sql/sql-server (Accessed: April 2026).

7. Patel, D., Gupta, M. and Singh, K. (2019) 'Digital Document Verification Module for University Administration Systems', International Journal of Advanced Research in Computer Science, Vol. 10, No. 4, pp. 201-208.

8. Pressman, R.S. and Maxim, B.R. (2020) Software Engineering: A Practitioner's Approach, 9th ed. McGraw Hill, New York.

9. Sommerville, I. (2016) Software Engineering, 10th ed. Pearson Education, Boston.

10. W3Schools (2024) 'HTML, CSS and JavaScript Tutorials', Available at: https://www.w3schools.com (Accessed: April 2026).

APPENDIX: USER MANUAL

This user manual provides step-by-step instructions for operating the **ScholarPath** system. It is designed for three types of users: **Students, Verifiers, and Administrators**.

System Requirements

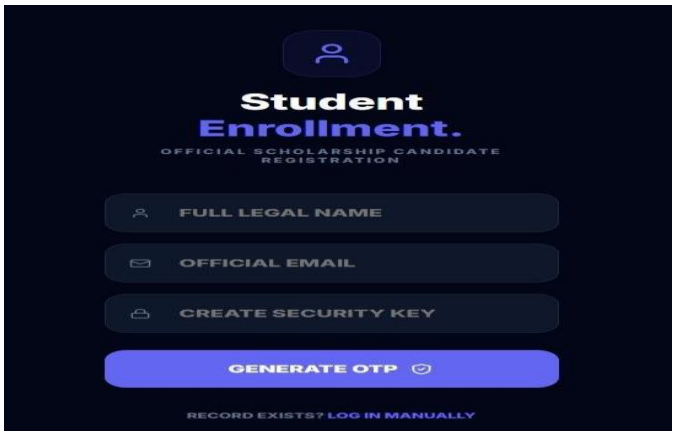
To access and use the system effectively, the following requirements must be met:

- **Operating System:** Windows 10 or later / macOS / Linux
- **Web Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge (latest versions recommended)
- **Internet Connection:** Stable internet connection (minimum 2 Mbps)
- **Screen Resolution:** 1024 × 768 pixels or higher

Student Instructions

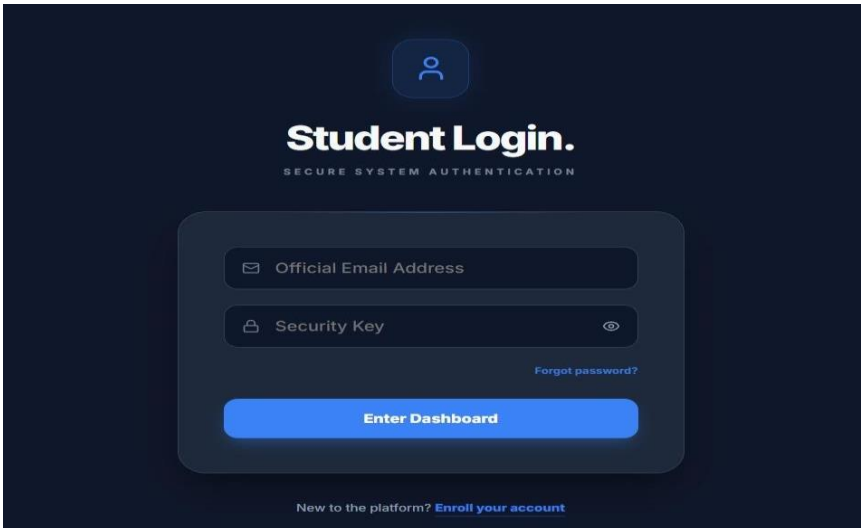
1. Registration (New User)

1. On the home page, click the **Register** button.
2. Enter the required details:
    a. Full Name
    b. Email Address
    c. Password
3. Confirm your password.
4. Click **Register**.
5. If the details are valid, your account will be created successfully.



2. Login

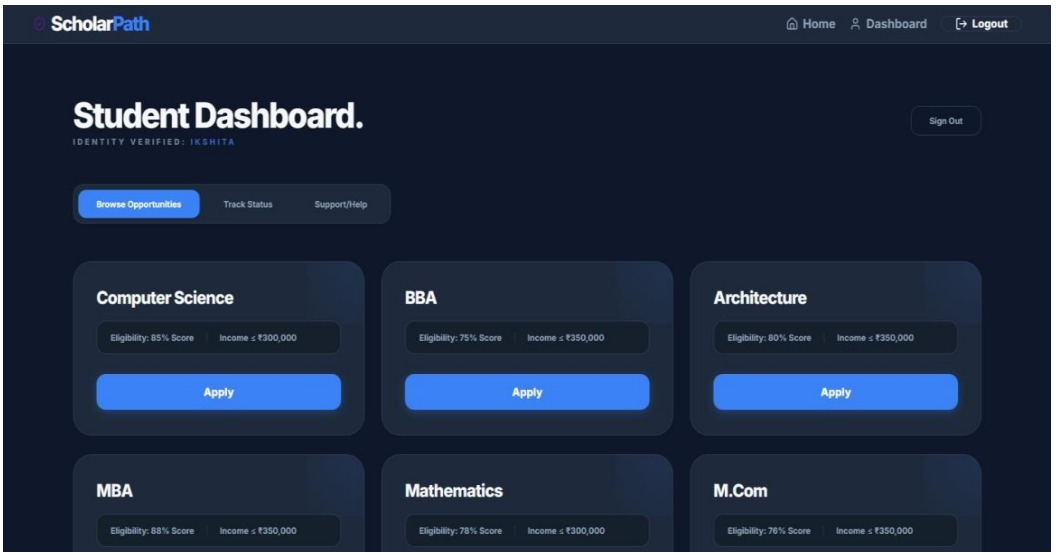
1. Click the **Login** button.
2. Enter your registered email and password.
3. Click **Sign In**.
4. Upon successful login, you will be redirected to the dashboard.



3. Dashboard Features

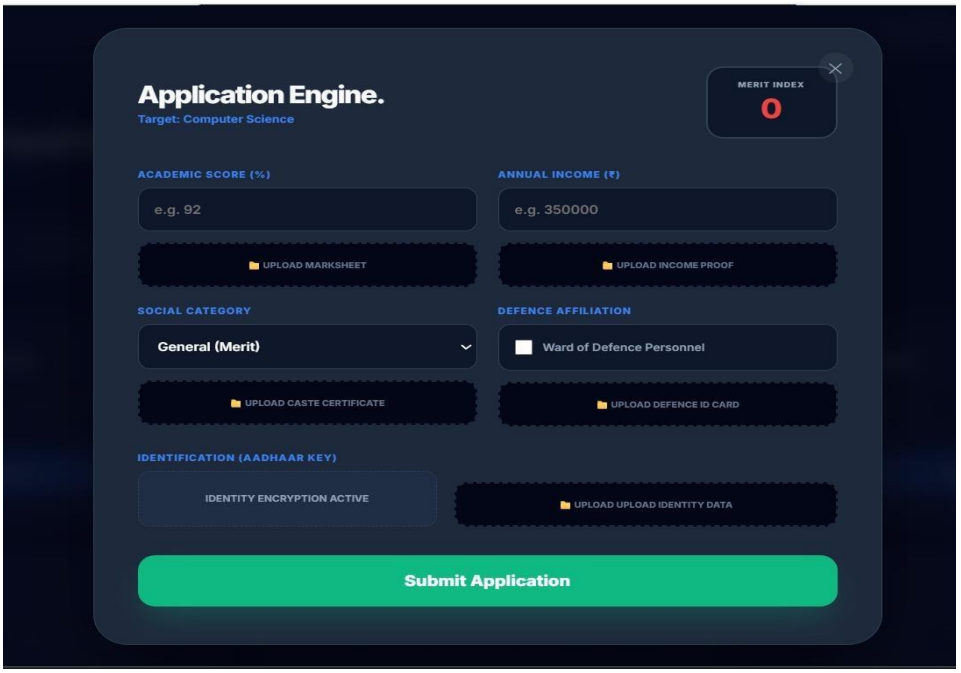
After logging in, the student dashboard provides the following options:

- **Opportunities:** View available scholarships and apply
- **Track Status:** Monitor the status of submitted applications
- **Support / Help:** Raise queries or submit help requests



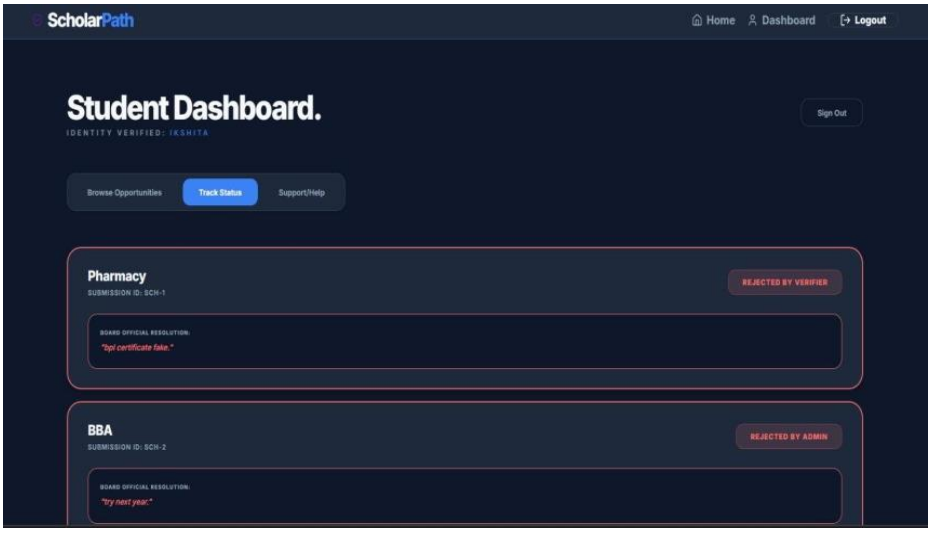
4. Applying for a Scholarship

1. Navigate to **Opportunities**.
2. Select a scholarship.
3. Fill in the application form with:
    a. Academic details
    b. Family income
    c. Other required information
4. Upload necessary documents (mark sheets, income proof, etc.).
5. Click **Submit**.
6. The application will be stored with status **Pending**.



5. Tracking Application Status

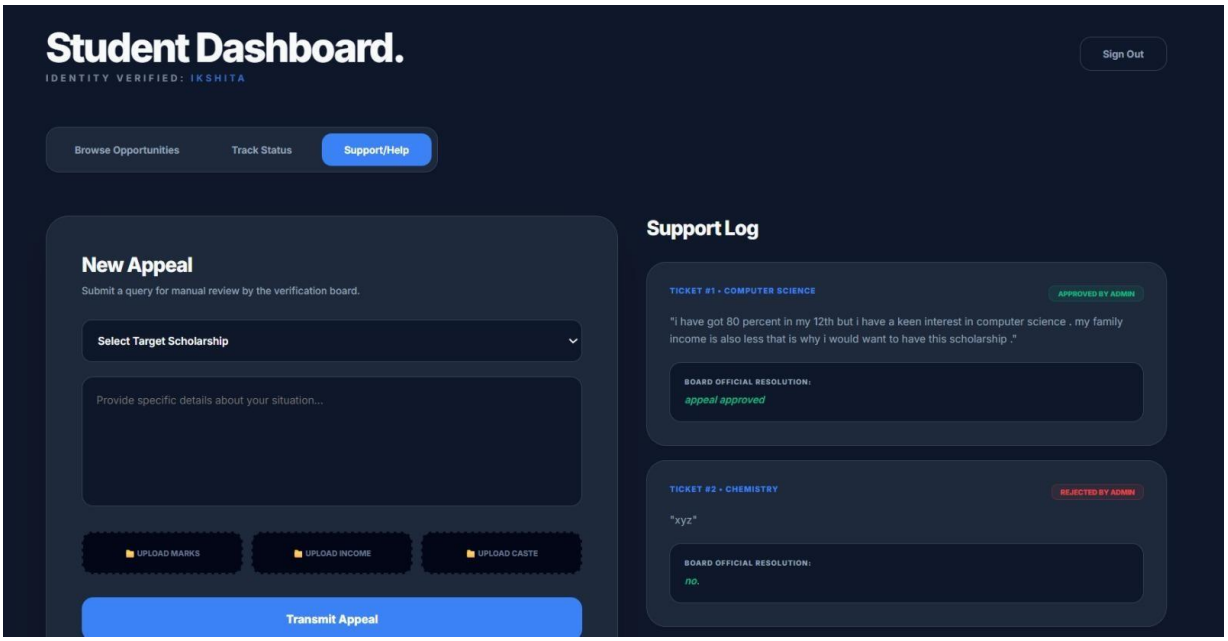
1. Go to **Track Status**.
2. View the current status:
    a. Pending
    b. Verified
    c. Approved
    d. Rejected



6. Requesting Help

1. Go to **Support / Help** section.
2. Enter the issue or reason.
3. Upload supporting documents if required.
4. Submit the request.
5. The request will be forwarded for review.

(You can continue next sections for Verifier and Admin if needed)



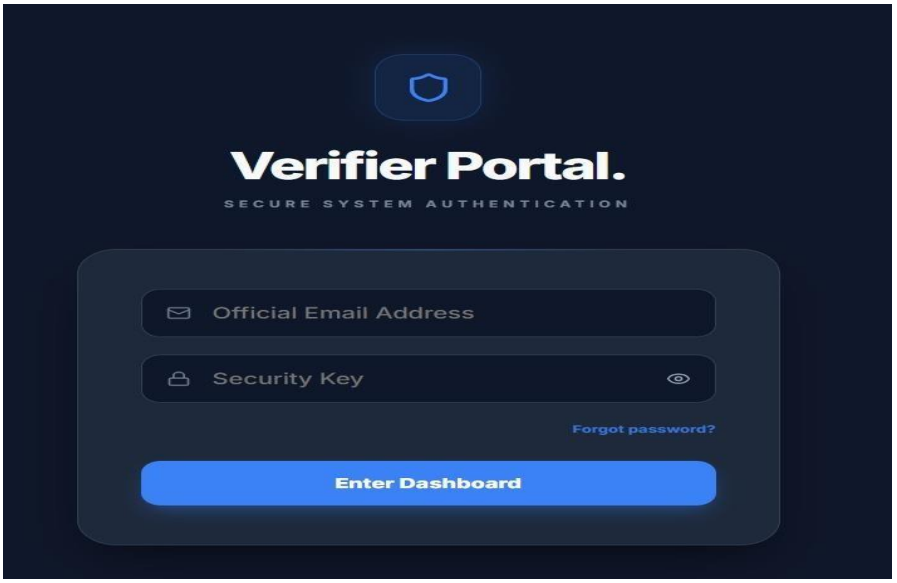
Verifier Instructions

The Verifier plays a crucial role in validating student applications and handling support requests before they are forwarded to the Administrator.

1. Verifier Login

1. Open the ScholarPath system in a web browser.
2. Click on the **Login** button.
3. Enter your registered **email** and **password**.
4. Click **Sign In**.
5. After successful authentication, you will be redirected to the Verifier Dashboard.

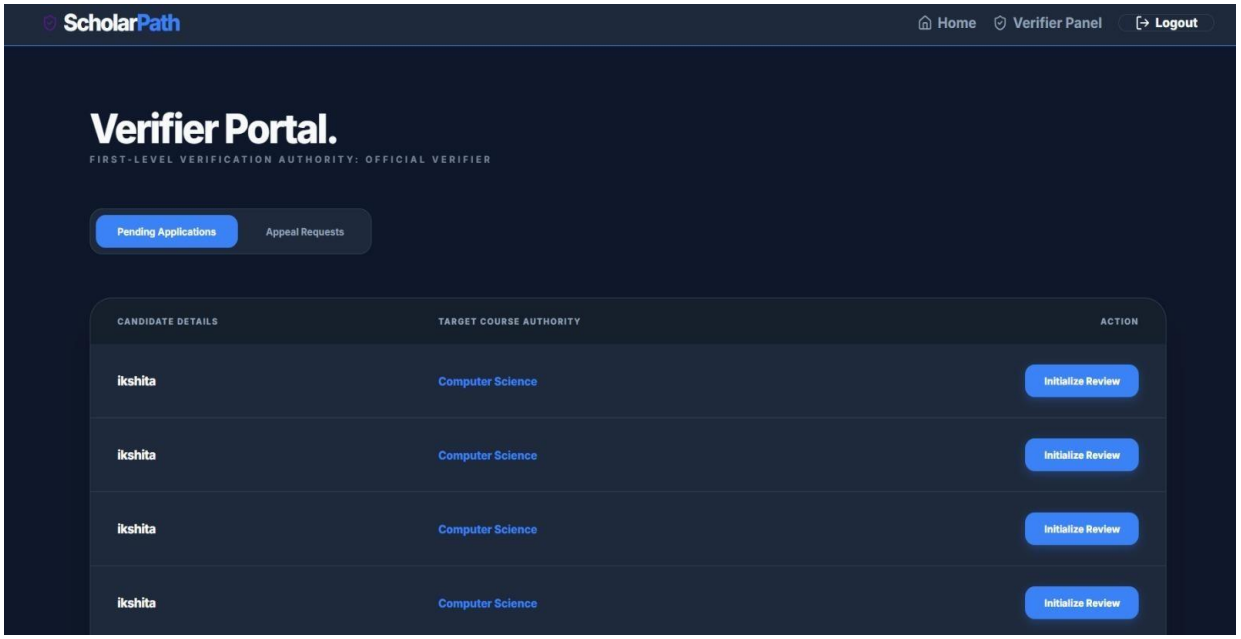
Note: Verifier access is granted only after approval by the Administrator.



2. Dashboard Features

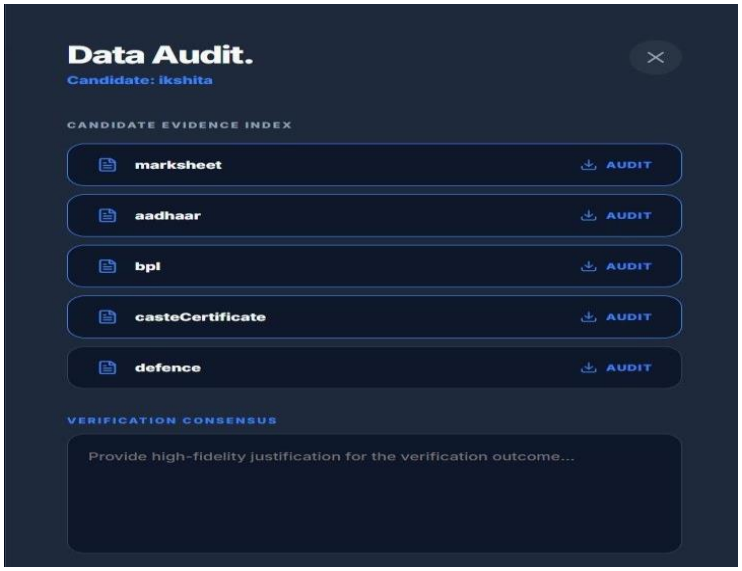
After logging in, the Verifier dashboard provides the following options:

- **Pending Applications:** View all student applications awaiting verification
- **Help Requests (Appeals):** Review student support or escalation requests



3. Reviewing Applications

1. Navigate to **Pending Applications**.
2. Select an application to review.
3. Check all submitted details:
    a. Academic information
    b. Income details
    c. Uploaded documents
4. Verify the authenticity and completeness of the data.



4. Updating Application Status

After reviewing the application:

- If the application is valid:

→ Mark it as **Verified** (forwarded to Admin)

- If the application is invalid:

→ Mark it as **Rejected**

→ Provide appropriate **remarks** explaining the reason

5. Submit the decision.
6. The updated status will be stored and visible to the student and admin.

5. Handling Help Requests (Appeals)

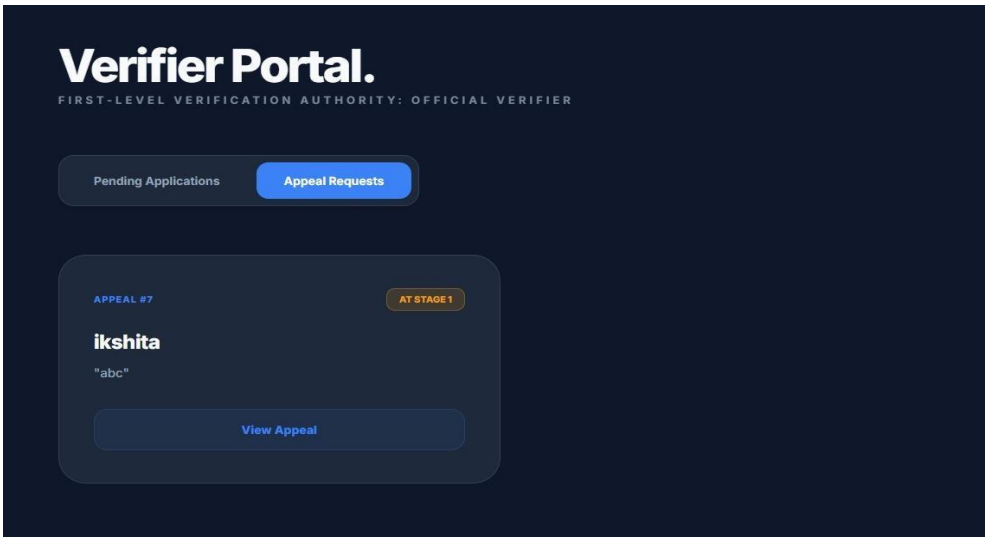
1. Navigate to **Help Requests**.
2. Select a request submitted by a student.
3. Review the issue and any attached documents.

- If the request is valid:

→ Accept and forward it to the Administrator with remarks

- If invalid:

→ Reject the request and provide remarks to the student



6. Logout

1. Click on the **Logout** button.
2. The session will be securely terminated.

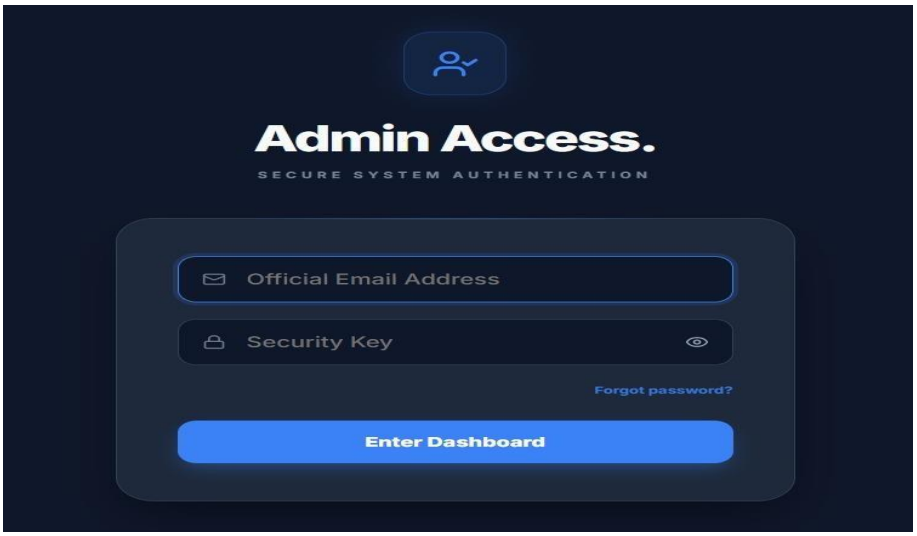
The Verifier ensures that only valid and complete applications proceed to the final decision stage, maintaining the integrity and reliability of the system.

Administrator Instructions

The Administrator has the highest level of control in the ScholarPath system and is responsible for final decision-making, scholarship management, and handling escalated requests.

1. Admin Login

1. Open the ScholarPath system in a web browser.
2. Click on the **Login** button.
3. Enter your registered **email** and **password**.
4. Click **Sign In**.
5. Upon successful login, you will be redirected to the Admin Dashboard.

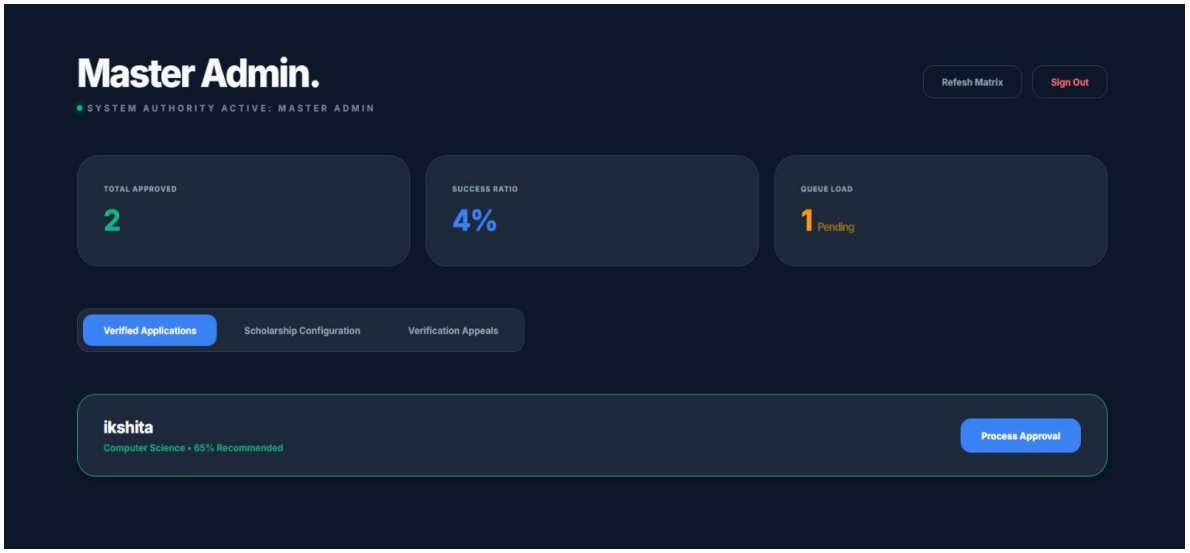


2. Dashboard Features

The Admin dashboard provides the following functionalities:

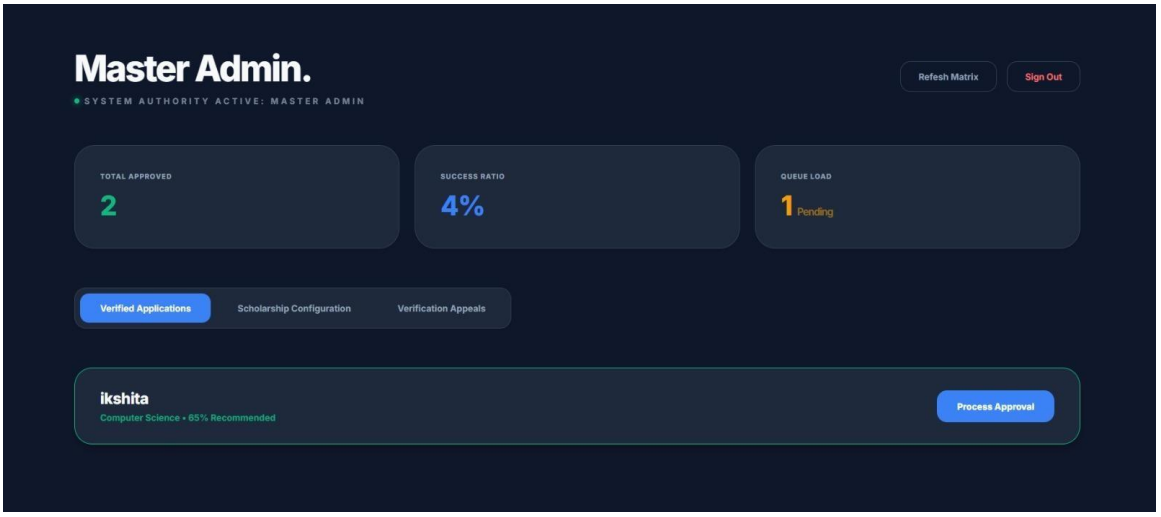
- **Verified Applications:** View applications approved by the Verifier
- **All Applications:** View all submitted applications

- **Scholarship Management:** Create, update, or delete scholarship opportunities
- **Help Requests (Appeals):** Handle escalated support requests



3. Reviewing Verified Applications

1. Navigate to **Verified Applications**.
2. Select an application.
3. Review all details and verifier remarks.



4. Final Decision on Applications

After reviewing:

- If the application meets requirements:

→ Mark as **Approved**

- If the application does not meet requirements:

→ Mark as **Rejected**

→ Provide appropriate **remarks**

4. Submit the decision.
5. The final status is updated and visible to the student.

5. Managing Scholarships

Create Scholarship:

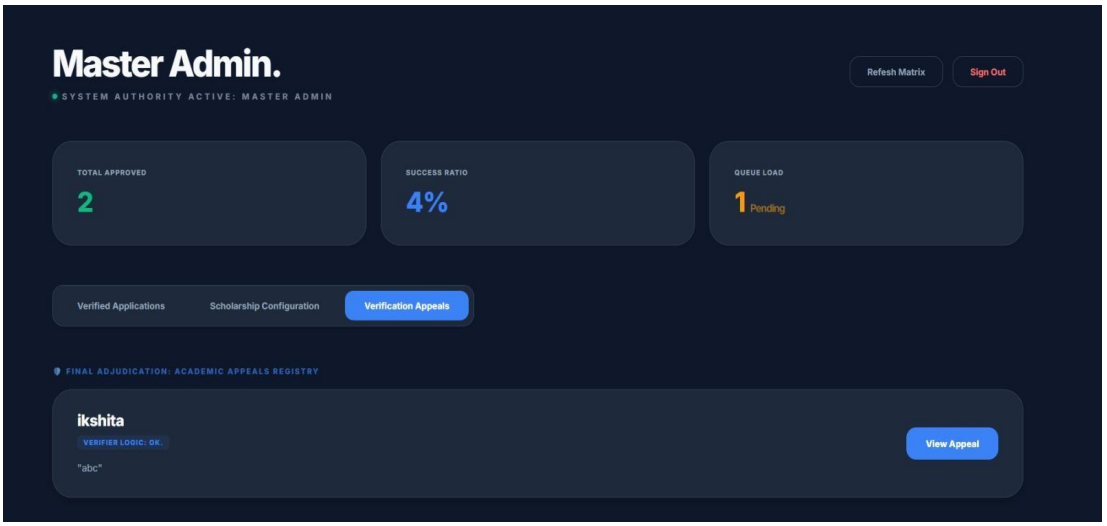
1. Go to **Scholarship Management**.
2. Click **Add New Scholarship**.
3. Enter required details (name, criteria, etc.).
4. Save the scholarship.

Update Scholarship:

1. Select an existing scholarship.
2. Modify the required details.
3. Save changes.

Delete Scholarship:

1. Select the scholarship.
2. Click **Delete**.
3. Confirm the action.



6. Handling Help Requests (Verification Appeals)

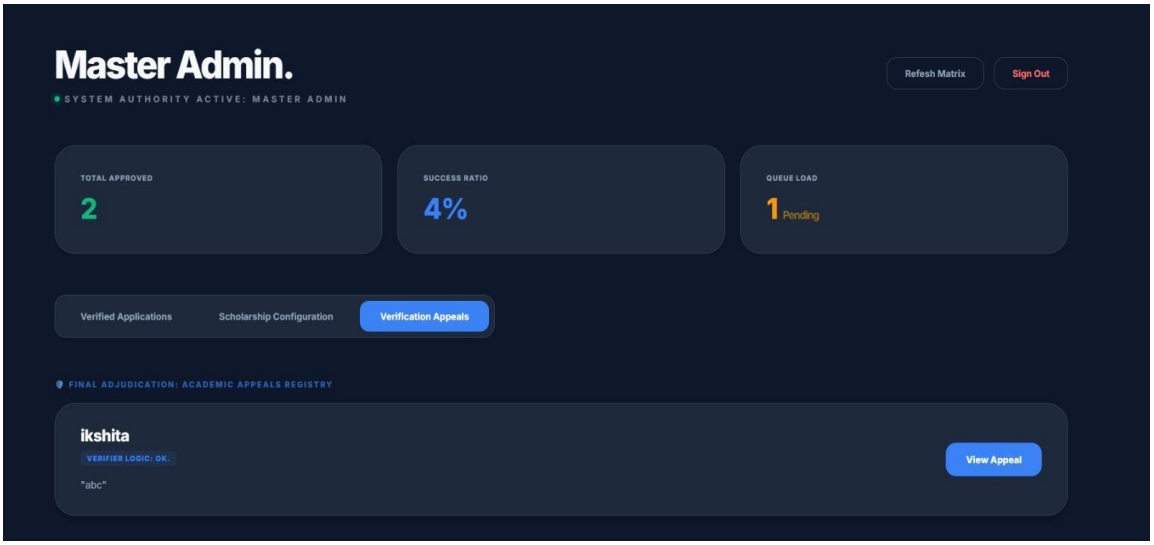
1. Navigate to **Help Requests**.
2. Review requests forwarded by the Verifier.
3. Analyze the issue and supporting documents.

- If valid:

→ Accept and resolve the request

- If not valid:

→ Reject the request with remarks



7. Logout

1. Click on the **Logout** button.
2. The session will be securely terminated.

The Administrator ensures final validation, manages scholarship data, and maintains overall system integrity and decision accuracy.